

WORKFORCE DOWNSIZING

ANALYSIS:

A DATA DRIVEN APPROACH WITH SQL

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PROJECT OBJECTIVE

The purpose of this project is to demonstrate my ability to work with real-world datasets by cleaning, transforming, and analyzing a sample data on global layoff using SQL. This project involves preparing the data for analysis through data cleaning techniques and writing SQL queries to uncover trends and insights related to company layoffs, such as industry impact, funding stages, year average layoffs and geographic distribution.

In a nut shell, the project showcases my proficiency in data cleaning, SQL querying, and analytical thinking. It is intended to serve as a strong portfolio piece for roles in data analysis and business intelligence.

TOOLS USED

- **SQL Server Management Studio (SSMS):** SSMS was used for the core of the project including data cleaning, manipulation, and querying. All major transformations and analysis were performed using SQL.
- **Microsoft Excel:** Excel was used for the initial review of the dataset and minor cleaning tasks such as identifying obvious inconsistencies and formatting issues before importing the data into SQL.
- **Kaggle:** The raw layoff dataset was sourced from Kaggle, a popular platform for data science datasets and competitions. The dataset contains information on company layoffs across various industries, countries, funding stages and funds raised.

DATASET DESCRIPTION

This project uses a publicly available layoff dataset sourced from Kaggle. The dataset captures detailed information on global company layoffs from 2020 to 2023, providing a strong foundation for analyzing workforce reduction trends across industries, countries, and company growth stages. It consists of 9 columns which are described as follows;

- **Company:** The name of the organization that carried out the layoff. This field helps identify which companies were most affected or active during periods of layoffs.
- **Location:** The city or region where the company is based or where the layoff was reported. This allows for geographical analysis of layoff patterns at a more granular level than country alone.
- **Industry:** The sector in which the company operates (e.g., Fintech, Health, Retail, Transportation, Education, etc.). This field is essential for evaluating which industries were hit hardest during certain time frames.
- **Total Laid Off:** The total number of employees laid off by a company. This column is numerical and helps quantify the impact of layoffs at the company level.
- **Percentage Laid Off:** This indicates the proportion of a company's workforce that was laid off. This is useful for understanding relative impact, especially in cases where total employee numbers vary widely between companies.
- **Date:** The specific date when the layoff occurred or was reported.
- **Stage:** The company's funding stage at the time of the layoff. This column provides insight into the company's financial maturity at the time of the layoff. (e.g, Seed, Series A, Acquired, Private Equity, etc.)
- **Country:** The country where the layoff occurred or where the company is headquartered. This enables a high-level view of how layoffs are distributed globally.
- **Funds Raised:** The total amount of capital the company had raised prior to the layoff, reported in millions. This can provide context on whether layoffs were a result of funding shortages or other strategic decisions.

PART ONE: DATA CLEANING PROCESS

i. **Creating DEMO table for Cleaning & Analysis:**

Before performing any transformations or analysis, a demo table was created to preserve the integrity of the original dataset. This ensures that any cleaning or testing performed does not alter the raw data.

```
-- Creating a demo database so as to maintain original data
```

```
select * into layoff_demo from layoffs
```

```
SELECT *  
FROM layoff_demo;
```

This SQL statement creates a new table named layoff_demo by copying all records and structure from the original layoffs table. The purpose of this is to protect the original data while allowing experimentation, data cleaning, and testing of queries in a separate, isolated environment.

	company	location	industry	total_laid_off	percentage_laid_off	date	stage
1	Atlassian	Sydney	Other	500	0.05	2023-03-06	Post-IPO
2	SiriusXM	New York City	Media	475	0.08	2023-03-06	Post-IPO
3	Alerzo	Ibadan	Retail	400	NULL	2023-03-06	Series B
4	UpGrad	Mumbai	Education	120	NULL	2023-03-06	Unknown
5	Loft	Sao Paulo	Real Estate	340	0.15	2023-03-03	Unknown

ii. Removing Duplicates:

It is crucial to identify duplicates before deleting them so as to be sure that the data is actually a duplicate. The query below was executed to detect duplicate rows in the layoff_demo table.

```
SELECT  
company, "location", industry, total_laid_off, percentage_laid_off, "date", stage,  
country, funds_raised_millions, count(*) as cnt  
from layoff_demo  
group by  
company, "location", industry, total_laid_off, percentage_laid_off, "date", stage,  
country, funds_raised_millions  
having count(*) > 1
```

Result:

	company	location	industry	total_laid_off	funds_raised_millions	cnt
1	Casper	New York City	Retail	NULL	339	2
2	Cazoo	London	Transportation	750	2000	2
3	Hibob	Tel Aviv	HR	70	45	2
4	Wildlife Studios	Sao Paulo	Consumer	300	260	2
5	Yahoo	SF Bay Area	Consumer	1600	6	2

Query to remove duplicates;

```
WITH duplicate_cte AS
(
    SELECT *,
    ROW_NUMBER() OVER(
    PARTITION BY company,"location",
    industry, total_laid_off, percentage_laid_off,"date",
    stage, country, funds_raised_millions
    ORDER BY company ASC)
    as row_num
    FROM layoff_demo
)

DELETE FROM duplicate_cte
WHERE row_num > 1;
```

iii. Standardizing Columns:

Here, using the TRIM() function I was able to remove extra spaces from the beginning and end of text values. This is to ensure consistent naming and prevent false distinctions when grouping, filtering, or joining on company names.

```
-- Standardizing Data

SELECT company, TRIM(company)
FROM layoff_demo;

-- Using TRIM function to remove any unnecessary space in the company column

UPDATE layoff_demo
SET company = TRIM(company);
```

iv. Correcting Industry Classification:

An inconsistent labeling was found in the industry column, some crypto-related companies were tagged as "Crypto" while others were labeled "Cryptocurrency". Hence, the query below was executed to correct this error;

```
-- The data in Layoff_demo labeled some companies as part of the 'Crypto' &
'Cryptocurrency' Industry
-- That's a clear error in the industry classification, all companies related to
Crypto should be under the Crypto Industry
-- To correct this, the following query is executed

UPDATE layoff_demo
SET industry = 'Crypto'
WHERE industry LIKE 'Crypto%';
```

v. Fixing Country Name Inconsistencies:

The country column had duplicate entries for United States due to a trailing period in some rows. Some rows displayed "United States." while some displayed "United States". To correct this, the following query was executed.

```
-- United States was displayed twice under the country column. To correct this,

SELECT DISTINCT country, TRIM(TRAILING '.' FROM country)
FROM layoff_demo
ORDER BY 1;

UPDATE layoff_demo
SET country = TRIM(TRAILING '.' FROM country)
WHERE country LIKE 'United States%';
```

vi. Reviewing Column Data Types:

To ensure each column was properly formatted for analysis, the data types were reviewed using:

```
-- To check Data Types

SELECT *
FROM INFORMATION_SCHEMA.COLUMNS
```

This helped confirm that key fields like date, total_laid_off, and funds_raised_millions had the correct data types and guided any needed adjustments.

vii. Identifying Missing Industry Values:

To locate missing or empty entries in the industry column, the following query was used:

```
--Identifying Industry NULL values

SELECT *
FROM layoff_demo

WHERE industry IS NULL
OR industry = '';
```

This helped identify records where the industry information was either completely missing or left blank, ensuring they could be addressed or categorized appropriately during cleaning.

Results;

	company	location	industry	total_laid_off	percentage_laid_off
1	Airbnb	SF Bay Area	NULL	30	NULL
2	Bally's Interactive	Providence	NULL	NULL	0.150000005960464
3	Juul	SF Bay Area	NULL	400	0.300000011920929
4	Carvana	Phoenix	NULL	2500	0.119999997317791
5	Juul	SF Bay Area	NULL	900	0.300000011920929

viii. Correcting Missing Industry Values:

To fill in missing industry data, values were inferred based on existing entries for the same company using the following query:

```
-- Correcting Industry NULL values

SELECT *
FROM layoff_demo
WHERE company = 'Airbnb';

UPDATE a
SET a.industry = b.industry
FROM layoff_demo a
JOIN layoff_demo b
    ON a.company = b.company
WHERE (a.industry IS NULL OR a.industry = '')
AND b.industry IS NOT NULL
AND b.industry != '';
```

This update populates empty industry fields by matching companies that already have valid industry values elsewhere in the dataset so as to ensure consistency without introducing external assumptions.

ix. Removing Records with Missing Layoff Data:

Records where both total_laid_off and percentage_laid_off were missing were deleted to maintain the reliability of the dataset.

```
-- Deleting all data where total_laid_off and percentage_laid_off is NULL
-- This is because, the information regarding them could not be retrieved
-- Hence, leaving it could alter our decision or result significantly

SELECT *
FROM layoff_demo
```

```
WHERE total_laid_off IS NULL
AND percentage_laid_off IS NULL;

DELETE
FROM layoff_demo
WHERE total_laid_off IS NULL
AND percentage_laid_off IS NULL;
```

Since these records lacked critical information for layoff analysis, keeping them could distort trends and lead to inaccurate insights.

x. **Standardizing Percentage Format:**

To ensure consistency and accuracy in percentage values, the percentage_laid_off column was converted to a decimal format with two decimal places using this query;

```
UPDATE layoff_demo
SET percentage_laid_off = CAST(percentage_laid_off AS decimal (10,2));
```

PART TWO: SQL QUERIES FOR LAYOFFS ANALYSIS

I. **Global Total Layoffs By Year:**

This query helps identify the years with the highest number of layoffs, highlighting potential periods of economic downturn or industry shifts. To analyze yearly layoff trends, the total number of employees laid off was grouped and summed by year using this query;

```
-- Trend Analysis Over Time

--TOTAL LAYOFFS PER YEAR

SELECT YEAR("date") as "Year", SUM(total_laid_off) as total_laid_off_by_year
FROM layoff_demo
WHERE "date" IS NOT NULL
GROUP BY YEAR("date")
ORDER BY SUM(total_laid_off) DESC;
```

	Year	total_laid_off_by_year
1	2022	160661
2	2023	125677
3	2020	80998
4	2021	15823

KEY INSIGHTS: *2022 recorded the highest number of layoffs globally, signaling a widespread post-COVID market correction. It more than doubled the figures from 2020 and 2021, highlighting a sharp industry-wide contraction.*

II. Global Average Percentage of Layoffs By Year:

This query calculates the average percentage of workforce laid off each year to assess the severity of layoffs relative to company size.

-- AVERAGE PERCENTAGE LAYOFF PER YEAR

```
SELECT YEAR("date")as "Year",
CAST(AVG(percentage_laid_off)as decimal (10,2)) as percentage_laid_off_by_year
FROM layoff_demo
WHERE "date" IS NOT NULL
GROUP BY YEAR("date")
ORDER BY AVG(percentage_laid_off) DESC;
```

Unlike total layoffs, this highlights how deeply companies were affected each year on average shedding light on the proportional impact of layoffs beyond raw numbers. The result is as follows;

	Year	percentage_laid_off_by_year
1	2021	0.61
2	2020	0.31
3	2022	0.24
4	2023	0.21

KEY INSIGHTS: *The highest average percentage of layoffs occurred in 2023, suggesting that although fewer companies may have laid off employees, those that did made deeper workforce cuts compared to earlier years.*

III. Yearly Global Layoff Trend: Pre-(Seed Companies) vs Post-IPO Companies:

This query examines the total number of employees laid off each year, comparing early-stage companies (typically Seed Companies) with later-stage companies (Post-IPO). Here's the query;

```
-- YEARLY TREND BETWEEN PRE- & POST - IPO COMPANIES - percentage_laid_off

SELECT YEAR("date")as "Year",
CAST(AVG(percentage_laid_off)as decimal (10,2)) as
    avg_percentage_laid_off_by_year, stage
FROM layoff_demo
WHERE "date" IS NOT NULL AND stage = 'Post-IPO' OR stage = 'Seed'
GROUP BY YEAR("date"), stage
ORDER BY YEAR("date") DESC;
```

Result;

	Year	avg_percentage_laid_off_by_year	stage
1	2023	0.14	Post-IPO
2	2023	0.65	Seed
3	2022	0.16	Post-IPO
4	2022	0.72	Seed
5	2021	0.16	Post-IPO
6	2020	0.18	Post-IPO
7	2020	0.70	Seed

KEY INSIGHTS:

Post-IPO companies had consistently higher total layoffs each year compared to Seed-stage firms, likely due to their larger workforce. However, this does not account for layoff severity (percentage laid off), which was often higher in early-stage companies.

IV. Industry-Based Layoff Trends (2020–2023):

These queries analyze which industries recorded the highest number of layoffs across individual years from 2020 to 2023. They check which industry recorded the highest total number of layoffs in 2020. In overall, it analyzes Global layoff totals per industry for the year 2022 to show peak sectors during COVID & post-COVID economic shifts.

```

SELECT
    industry,
    SUM(CASE WHEN YEAR([date]) = 2020 THEN total_laid_off ELSE 0 END) AS layoffs_2020,
    SUM(CASE WHEN YEAR([date]) = 2021 THEN total_laid_off ELSE 0 END) AS layoffs_2021,
    SUM(CASE WHEN YEAR([date]) = 2022 THEN total_laid_off ELSE 0 END) AS layoffs_2022,
    SUM(CASE WHEN YEAR([date]) = 2023 THEN total_laid_off ELSE 0 END) AS layoffs_2023,
    SUM(total_laid_off) AS total_layoffs_2020_2023
FROM layoff_demo
WHERE [date] IS NOT NULL AND industry IS NOT NULL
GROUP BY industry
ORDER BY total_layoffs_2020_2023 DESC;

```

	industry	layoffs_2020	layoffs_2021	layoffs_2022	layoffs_2023	total_layoffs_2020_2023
1	Consumer	5264	3600	19726	15663	44253
2	Retail	8002	1088	20644	13609	43343
3	Other	466	515	6346	28512	35839
4	Transportation	14656	200	15227	3648	33731
5	Finance	8624	0	12674	7053	28351
6	Healthcare	1125	0	15058	9770	25953
7	Food	6218	2644	11288	2705	22855
8	Real Estate	3699	2900	10007	1034	17640
9	Travel	13983	0	1437	1539	16959
10	Hardware	0	0	605	13223	13828
11	Education	685	1943	8728	1982	13338
12	Sales	1334	0	2601	9311	13246
13	Crypto	91	0	8263	2339	10693
14	Marketing	2827	21	3707	3682	10327

V. Top 5 Most Impacted Industries by Total Layoffs and Avg. % (2020–2023):

This ranks the top 5 industries with the most layoffs overall, while also showing the average percentage laid off in each industry. The query is as follows;

```

-- TOP 5 MOST IMPACTED INDUSTRIES BY TOTAL AND PERCENTAGE FROM 2020-2023

SELECT TOP 5 industry,
    SUM(total_laid_off) as total_laid_off_by_industry
FROM layoff_demo
WHERE "date" IS NOT NULL AND industry IS NOT NULL
GROUP BY industry
ORDER BY SUM(total_laid_off) DESC ;

SELECT TOP 5 industry,
    CAST(AVG(percentage_laid_off) as decimal (10,2)) as avg_percentage_laid_off_by_year
FROM layoff_demo

```

	industry	total_laid_off_by_industry
1	Consumer	44253
2	Retail	43343
3	Other	35839
4	Transportation	33731
5	Finance	28351

<pre> SELECT TOP 5 industry, CAST(AVG(percentage_laid_off)as decimal (10,2)) as avg_percentage_laid_off_by_year FROM layoff_demo WHERE "date" IS NOT NULL AND industry IS NOT NULL GROUP BY industry ORDER BY avg_percentage_laid_off_by_year DESC ; </pre>		
100 %		
Results Messages		
	industry	avg_percentage_laid_off_by_year
1	Aerospace	0.56
2	Education	0.36
3	Travel	0.35
4	Food	0.32
5	Real Estate	0.32

KEY INSIGHTS:

- The Consumer, Retail, and Other industries had the highest total layoffs, with Consumer leading. However, these were not the most severe when measured by percentage laid off, suggesting layoffs were more distributed across larger companies.
- Aerospace, Education, and Travel had the highest average layoff percentages, indicating that when layoffs occurred in these sectors, they were more severe relative to company size.

VI. Yearly Average Layoffs per Industry:

This query was executed to display the 3-year average of total layoffs for each industry, helping to identify consistently impacted sectors.

```

-- AVERAGE LAYOFF PER YEAR FROM EACH INDUSTRY

SELECT industry,
SUM(total_laid_off)/3 as Yearly_avg_total_laid_off_by_industry
FROM layoff_demo
WHERE "date" IS NOT NULL AND industry IS NOT NULL
GROUP BY industry
ORDER BY SUM(total_laid_off)/3 DESC;

```

KEY INSIGHTS:

2020: The Consumer industry experienced the highest layoffs in 2020, likely driven by the immediate economic disruption of COVID-19, which hit non-essential goods and services the hardest.

2021: Transportation and Retail sectors topped layoffs in 2021, reflecting lingering supply chain disruptions and shifts in consumer behavior.

2022: Tech emerged as the most affected industry in 2022, with massive cuts in startups and public companies due to over hiring during the pandemic and tightening investor sentiment.

2023: While layoffs declined slightly in volume, Retail and Education saw significant cuts, possibly reflecting a reallocation of workforce needs and tighter operational budgeting post-pandemic.

	industry	total_laid_off_by_industry_2022			
1	Retail	20644	16	Media	1879
2	Consumer	19726	17	Logistics	1473
3	Transportation	15227	18	Travel	1437
4	Healthcare	15058	19	Support	1386
5	Finance	12674	20	Data	1314
6	Food	11288	21	Product	861
7	Real Estate	10007	22	HR	738
8	Education	8728	23	Energy	635
9	Crypto	8263	24	Hardware	605
10	Other	6346	25	Construction	433
11	Fitness	5546	26	Recruiting	380
12	Infrastructure	4916	27	Legal	350
13	Marketing	3797	28	Fin-Tech	150
14	Security	3699	29	Aerospace	100
15	Sales	2601			

VII. Total Layoffs by Country:

This query was executed to rank countries based on the total number of layoffs recorded from 2020–2023.

```
-- TOTAL LAYOFFS PER COUNTRY

SELECT country, SUM(total_laid_off) as total_laid_off_by_country
FROM layoff_demo
WHERE country IS NOT NULL
GROUP BY country
ORDER BY SUM(total_laid_off) DESC ;
```

KEY INSIGHTS:

The United States accounted for the highest total layoffs globally by a significant margin followed by India and Canada. This reflects both the large tech presence in these countries and the transparency of reporting.

	country	total_laid_off_by_country
1	United States	256559
2	India	35993
3	Netherlands	17220
4	Sweden	11264
5	Brazil	10391
6	Germany	8701
7	United Kingdom	6398
8	Canada	6319
9	Singapore	5995
10	China	5905
11	Israel	3638
12	Indonesia	3521
13	Australia	2324
14	Nigeria	1882
15	United Arab Emirates	995
16	France	915
17	Hong Kong	730
18	Austria	570

19	Russia	400
20	Kenya	349
21	Estonia	333
22	Argentina	323
23	Senegal	300
24	Mexico	270
25	Ireland	257
26	Finland	250
27	Spain	250
28	Denmark	240
29	Myanmar	200
30	Norway	140
31	Colombia	130
32	Bulgaria	120
33	Portugal	115
34	Malaysia	100
35	Japan	85

36	Romania	80
37	Seychelles	75
38	Switzerland	62
39	Lithuania	60
40	Thailand	55
41	New Zealand	45
42	Luxembourg	45
43	Chile	30
44	Poland	25

VIII. Average Percentage Layoffs by Country:

This query was executed to show countries with the highest average layoff percentages, indicating where layoffs had more severe effects relative to company size.

```
-- AVG PERCENTAGE LAY OFFS PER COUNTRY
```

```
SELECT country,
CAST(AVG(percentage_laid_off)as decimal (10,2)) as avg_percentage_laid_off_by_year
FROM layoff_demo
WHERE country IS NOT NULL
GROUP BY country
ORDER BY avg_percentage_laid_off_by_year DESC ;
```

Result;

	country	avg_percentage_laid_off_by_year			
1	Vietnam	0.83	16	Italy	0.30
2	Denmark	0.68	17	Nigeria	0.29
3	Pakistan	0.54	18	Kenya	0.29
4	Egypt	0.50	19	France	0.28
5	Switzerland	0.50	20	Singapore	0.27
6	United Arab Emirates	0.46	21	Austria	0.27
7	Russia	0.40	22	Mexico	0.27
8	Australia	0.39	23	Germany	0.26
9	Portugal	0.35	24	United States	0.25
10	Myanmar	0.33	25	Israel	0.24
11	Indonesia	0.33	26	Argentina	0.23
12	India	0.32	27	Canada	0.22
13	Seychelles	0.32	28	Brazil	0.20
14	Chile	0.30	29	South Korea	0.20
15	United Kingdom	0.30	30	Ireland	0.19
36	Sweden	0.15			
37	Hungary	0.14			
38	Malaysia	0.14			
39	Turkey	0.14			
40	Hong Kong	0.13			
41	Norway	0.12	31	China	0.18
42	Luxembourg	0.11	32	Netherlands	0.18
43	Spain	0.10	33	Estonia	0.16
44	Bulgaria	0.10	34	Senegal	0.15
45	Thailand	0.08	35	Finland	0.15
46	Colombia	0.03			

KEY INSIGHTS: *Countries like Singapore, Sweden, and Netherlands had higher average layoff percentages, indicating that when layoffs occurred, they were more severe — often affecting a large portion of company staff.*

IX. Layoffs by Industry and Country (Cross-Dimensional Analysis):

This query was executed to combine both industry and country to provide a two-dimensional view—revealing which industries were most impacted in each country.

```
-- CROSS DIMENSIONAL ANALYSIS  
-- LAYOFFS BY INDUSTRY AND COUNTRY
```

```
SELECT country, industry, SUM(total_laid_off) as total_laid_off_by_country  
FROM layoff_demo  
WHERE country IS NOT NULL  
GROUP BY country, industry  
ORDER BY country, total_laid_off_by_country DESC;
```

Results:

	country	industry	total_laid_off_by_country
1	Argentina	Crypto	180
2	Argentina	Finance	53
3	Argentina	Marketing	50
4	Argentina	Other	40
5	Australia	Other	508
6	Australia	Food	420
7	Australia	Crypto	264
8	Australia	Finance	243
9	Australia	Real Estate	199
10	Australia	Retail	199

KEY INSIGHTS: *The Tech industry in the U.S. saw the most significant number of layoffs, while Consumer layoffs were more regionally distributed. This cross-dimensional view helps pinpoint where industry-specific layoffs were most concentrated geographically.*

X. Global Total Layoffs By Company Stage From 2020-2023:

This query calculates the total number of layoffs recorded for each company funding stage (e.g., Seed, Series A, Private Equity, Post-IPO, Acquired, etc.). This is important in order to analyze layoffs by funding stage helps reveal which types of companies (early-stage, growth-stage, mature) were most affected by workforce reductions. Below is the result for the query;

```
SELECT stage,
       SUM(CASE WHEN YEAR([date]) = 2020 THEN total_laid_off ELSE 0 END) AS layoffs_2020,
       SUM(CASE WHEN YEAR([date]) = 2021 THEN total_laid_off ELSE 0 END) AS layoffs_2021,
       SUM(CASE WHEN YEAR([date]) = 2022 THEN total_laid_off ELSE 0 END) AS layoffs_2022,
       SUM(CASE WHEN YEAR([date]) = 2023 THEN total_laid_off ELSE 0 END) AS layoffs_2023,
       SUM(total_laid_off) AS total_layoffs_2020_2023_by_stage
FROM   layoff_demo
WHERE  [date] IS NOT NULL AND stage IS NOT NULL
GROUP BY stage
ORDER BY total_layoffs_2020_2023_by_stage DESC;
```

	stage	layoffs_2020	layoffs_2021	layoffs_2022	layoffs_2023	total_layoffs_2020_2023_by_stage
1	Post-IPO	22672	2895	79373	98602	123929
2	Unknown	7632	9097	19127	48630	48630
3	Acquired	12169	2984	5457	6966	27576
4	Series C	4149	85	13072	2711	20017
5	Series D	8837	236	8351	1801	19225
6	Series B	5609	86	7514	2102	15311
7	Series E	3807	0	6469	2421	12697
8	Series F	3967	323	4546	1096	9932
9	Private Equity	2369	0	3907	1681	7957
10	Series H	850	0	5539	855	7244
11	Series A	1723	117	3425	413	5678
12	Series G	2436	0	970	291	3697
13	Series J	1920	0	1450	200	3570
14	Series I	2645	0	160	50	2855
15	Seed	213	0	1269	154	1636
16	Subsidiary	0	0	0	1094	1094

KEY INSIGHTS:

- *Post-IPO companies: **123,929** — clearly the most impacted, likely due to public market volatility and pressure to reduce costs.*
- *Unknown stage: **48,630** — possibly smaller firms or unclassified data, still significantly affected.*
- *Acquired companies: **27,576** — layoffs likely driven by post-acquisition restructuring and integration.*

Among startups:

- *Series C–E companies saw between **12,697** and **20,017** total layoffs, indicating growth-stage companies were hit hard as funding dried up.*
- *Early-stage startups (Seed, Series A): fewer than **6,200** layoffs combined — suggesting limited impact due to smaller teams and lower burn rates.*

PROJECT LIMITATIONS

While the analysis offers valuable insights into global layoff patterns, it's important to acknowledge certain limitations in the dataset and methodology. These constraints may influence the accuracy, completeness, or generalizability of the findings, and should be considered when interpreting the results. The limitations are as follows;

- 1. Incomplete or Missing Data:** Some entries had missing values for critical columns like industry, total_laid_off, or percentage_laid_off, which could limit the accuracy of trend analysis despite cleaning efforts.
- 2. Static Dataset:** The dataset used was a snapshot and does not update in real-time. New layoffs or updates are not captured unless the dataset is refreshed from source.
- 3. Inconsistent Company Reporting:** Not all companies disclose layoffs publicly. As a result, the data may disproportionately represent larger or more visible organizations, skewing industry or country trends.
- 4. No Workforce Baseline Data:** Since the total employee count per company wasn't always available, some percentage calculations may not reflect actual severity.
- 5. Funding Stage Definitions Vary:** Company classification by funding stage (e.g., Seed vs. Series A) may differ by region or investor, potentially introducing inconsistency.

CONCLUSION

In conclusion, this project demonstrates the power of SQL in managing, cleaning, and analyzing real-world datasets. Through a comprehensive review of global layoff data from 2020 to 2023, key trends were uncovered in terms of industries affected, funding stage dynamics, and geographical impact.

By transforming raw data into actionable insights, this analysis highlights the critical role of data analysis in workforce planning and economic monitoring. The project also showcases my ability to approach complex data problems methodically, from cleaning and transformation to querying and interpretation which is a core strength for any data-driven role in today's business environment.